

Travelling Salesman Problem: NP - Hard Optimization Problem

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1 Problem Description

We consider the **optimization version** of the Travelling Salesman Problem (TSP), an NP-hard combinatorial optimization problem. Given cities $V = \{1, \dots, n\}$ with coordinates $p_i = (x_i, y_i) \in \mathbb{R}^2$, find a permutation $\pi = (\pi_1, \dots, \pi_n)$ minimizing

$$L(\pi) = \sum_{k=1}^{n-1} d(\pi_k, \pi_{k+1}) + d(\pi_n, \pi_1),$$

where

$$d(i, j) = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2},$$

subject to every city appearing exactly once. The corresponding decision problem, asking whether a tour of cost at most K exists, is NP-complete. Experiments use randomly generated Euclidean instances with $x_i, y_i \sim U(0, 100)$.

2 Prompting Strategies

We used three progressively structured prompting strategies to evaluate LLM-assisted algorithm generation and refinement.

1. **Zero-shot prompting:** The LLM was asked for a reliable classical TSP algorithm, its complexity, and an implementation. It generated the exact **Held–Karp** dynamic-programming algorithm, with $O(n^2 2^n)$ time and $O(n 2^n)$ space, which served as the ground-truth solver for small instances.
2. **Role + constraint prompting:** The LLM was instructed to act as an algorithms researcher and design a practical heuristic for medium-sized Euclidean TSP instances. It generated the classical **Nearest Neighbor + 2-Opt** heuristic, combining greedy construction with local improvement.
3. **Critique + refinement prompting:** The LLM was given the NN + 2-Opt algorithm and asked to identify structural weaknesses and propose an improvement. It identified deterministic construction and local-optimum trapping, and generated **GRASP + 2-Opt**, using randomized greedy construction followed by repeated 2-Opt search. GRASP and 2-Opt were treated as established techniques rather than novel algorithms.

All generated implementations were independently executed on identical instances, and returned tour lengths were recomputed from the tours. For small instances, solutions were compared with Held–Karp using

$$\text{Gap}(\%) = \frac{L - L^*}{L^*} \times 100.$$

n	NN+2-Opt	GRASP+2-Opt	Improvement
5–10	8/9 optimal	9/9 optimal	—
20	434.83	405.06	6.85%
50	601.77	566.56	5.85%

For the nine small instances, GRASP + 2-Opt matched the Held–Karp optimum on **9/9** instances, compared with **8/9** for NN + 2-Opt. On the controlled medium instances, GRASP reduced average tour length by **6.85%** at $n = 20$ and **5.85%** at $n = 50$, but required substantially higher computation. In an additional five-seed experiment at $n = 100$, GRASP + 2-Opt was worse on average than the deterministic baseline despite its higher runtime, showing that the generated refinement is not universally better.

The resulting

LLM generation → verification → evaluation → refinement

workflow also produces structured prompt–algorithm–evaluation records containing the prompting strategy, generated algorithm, verified implementation, instance, objective value, runtime, and refinement feedback, forming a prototype dataset for future LLM training and evaluation.

3 Conclusion

The experiments show that structured prompting can progressively guide an LLM from exact algorithm selection to heuristic generation and algorithmic refinement. The generated GRASP + 2-Opt heuristic improved solution quality on the tested small and medium instances, including a 9/9 match with the exact optimum, but incurred substantially higher computational cost and did not consistently outperform NN + 2-Opt at larger scale.

The central finding is therefore not that LLMs universally produce better algorithms, but that they can effectively assist **algorithm generation and refinement when coupled with independent verification and empirical evaluation**. The resulting prompt-to-evaluation records additionally provide a foundation for future research on LLM-based combinatorial algorithm design.

4 References

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